

PRACTICAL - 1

AIM :

Setting Up Android Development Environment on Linux Install Android Studio on your Linux machine. Create a new Android project and understand the project structure. Familiarize yourself with the Android Virtual Device (AVD) manager and create a virtual device.

THEORY :

Android Studio is the official Integrated Development Environment (IDE) for Android app development, developed by Google.

It provides tools for:

- Writing code (Kotlin/Java)
- Designing user interfaces
- Testing applications using emulator
- Debugging and performance analysis

Android Studio uses Gradle as its build system and includes features like:

- Code editor with auto-completion
- Layout editor for UI design
- Android Virtual Device (AVD) for testing apps

Installing Android Studio is the first step in setting up a complete Android development environment.

STEPS :

Step 1: Update System

Open terminal and run:

```
sudo apt update  
sudo apt upgrade
```

Step 2: Install Java (JDK 17)

Verify installation:

```
sudo apt install openjdk-17-jdk  
java -version
```

Step 3: Install Snap (if not available)

```
sudo apt install snapd
```

Step 4: Install Android Studio

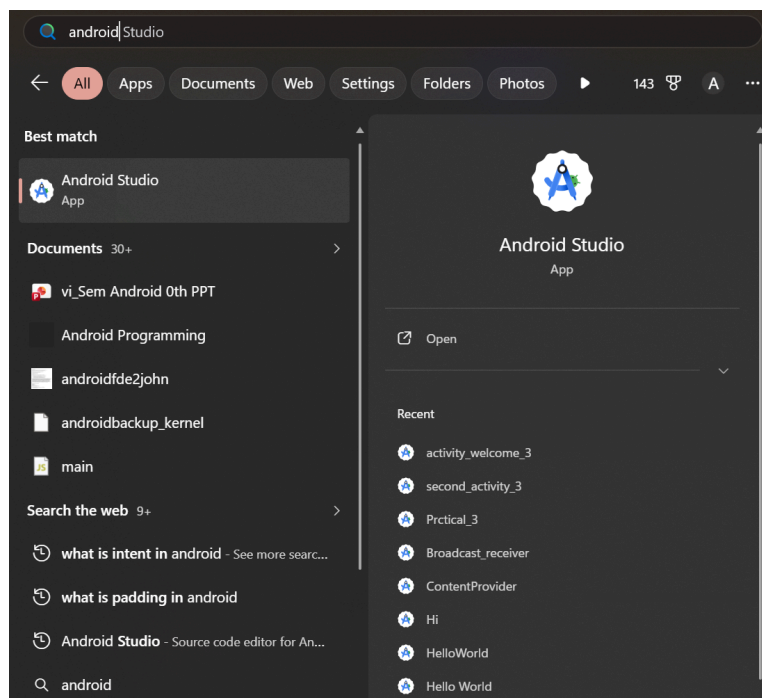
```
sudo snap install android-studio --classic
```

Step 5: Launch Android Studio

```
android-studio
```

Step 6: Complete Setup Wizard

- Select Standard Installation
- Choose theme (Light/Dark)
- Accept licenses
- Download required SDK tools



Android Project Structure :

When a new project is created, it contains organized folders and files:

- **app/** → Main module of the application
- **java/** → Contains source code (Kotlin/Java files like MainActivity)
- **res/** → Resources such as:
 - **layout/** → UI design (XML files)
 - **drawable/** → Images
 - **values/** → Colors, strings, styles
- **AndroidManifest.xml** → Defines app configuration, permissions, activities
- **Gradle Scripts** → Used for building and managing dependencies

This structure helps in maintaining clean and modular app development.

Android Virtual Device (AVD) :

AVD is an emulator provided by Android Studio that simulates a real Android device.

It allows developers to:

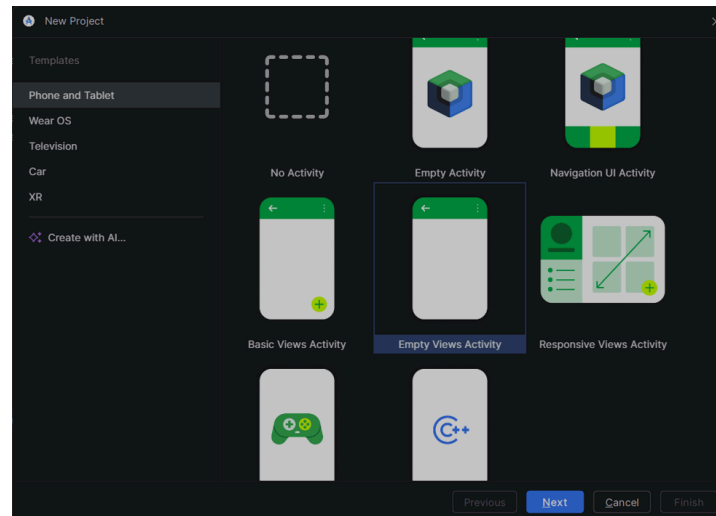
- Run and test apps without a physical device
- Simulate different screen sizes and Android versions
- Debug applications easily

AVD Manager is used to create and manage virtual devices.

STEPS :

Part A: Create a New Project

- a. Open Android Studio
- b. Click New Project
- c. Select Empty Views Activity
- d. Select Language → Java

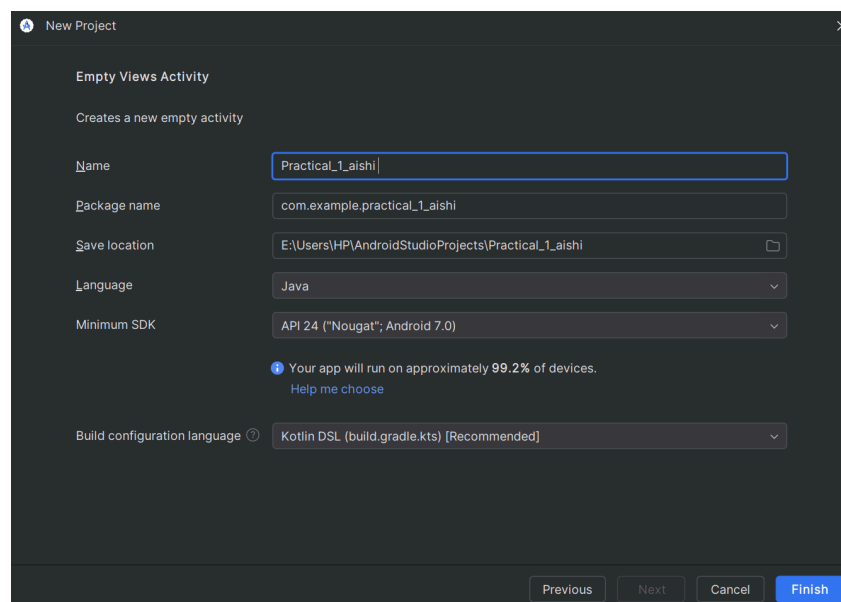


2. Configure Project

e. Enter Project Details:

- Name → MyFirstApp
- Package Name → com.example.myapp
- Minimum SDK → API 21 or above

f. Click Finish



3. Build the Project

g. Wait for Gradle build to complete

h. Project will open in editor window

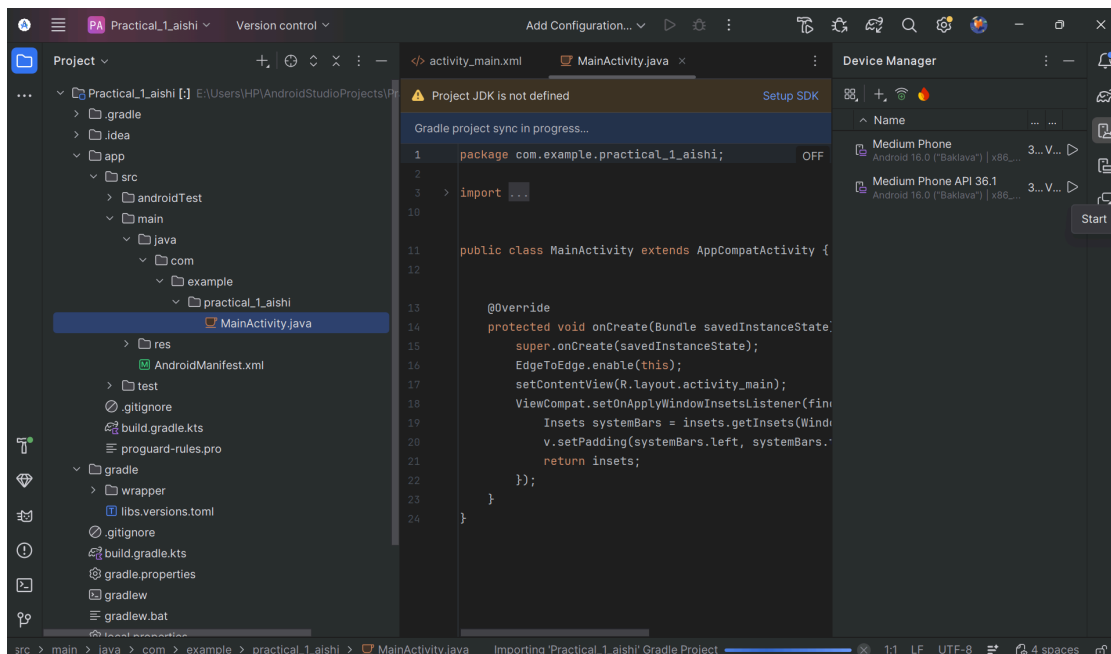
4. Understand Project Structure

i. Go to Project Panel

j. Select Android View

k. Observe the following:

- **app/java/** → Java source code (MainActivity.java)
- **res/layout/** → UI design files
- **res/drawable/** → Images
- **res/values/** → Strings, colors, styles
- **AndroidManifest.xml** → App configuration
- **Gradle Scripts** → Build files



Part B: Android Virtual Device (AVD) Setup

5. Open AVD Manager

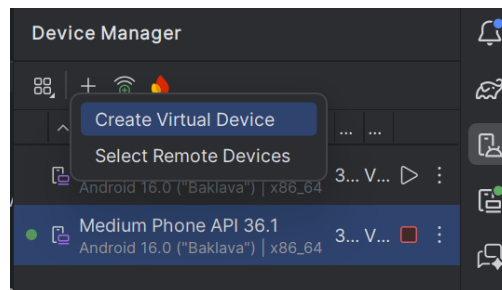
l. Click Device Manager (top right corner)

6. Create Virtual Device

m. Click Create Device

n. Select a device (e.g., Pixel 4)

o. Click Next

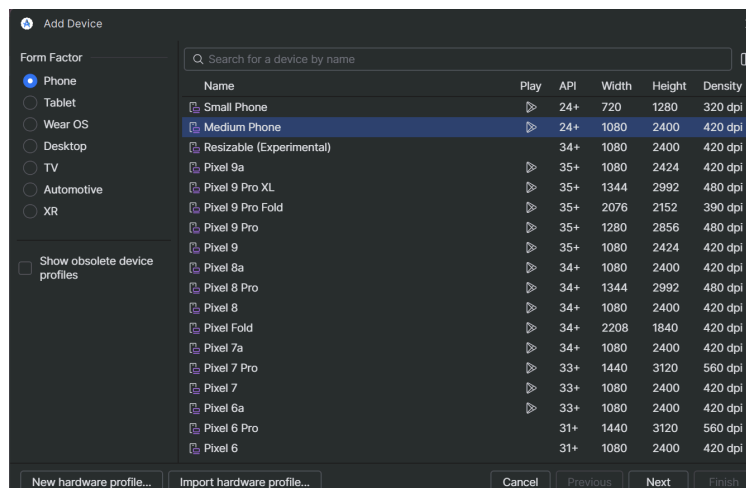


7. Select System Image

p. Choose Android version (latest recommended)

q. Click Download (if not installed)

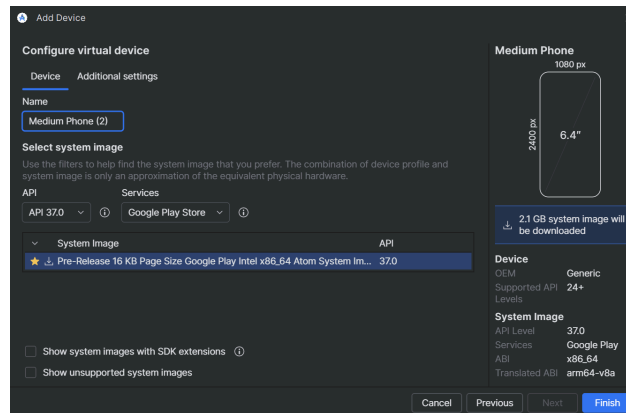
r. Click Next



8. Finish AVD Setup

s. Click Finish

t. Virtual device will be created

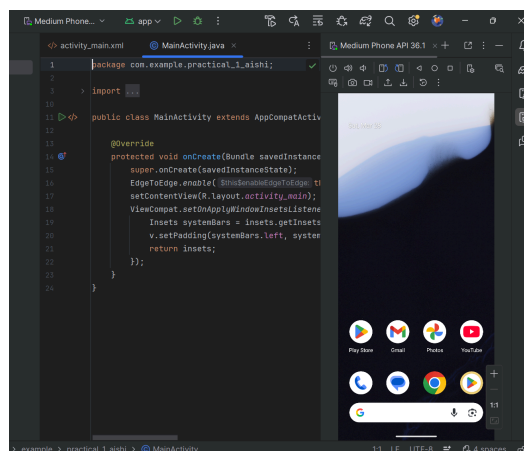


9. Run Emulator

u. Click Play button in Device Manager
v. Emulator will start

10. Run Application

w. Click Run
x. Select the virtual device
y. Application will launch on emulator



CONCLUSION :

A new Android project was successfully created using Android Studio with Java language. The project structure was understood, and an Android Virtual Device (AVD) was configured to run the application.